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Roll No

ME-7005(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power Technology

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define chain reaction. List out advantages and disadvantages of Nuclear Power Station.
b) What are the reasons for using cadmium as a control rod in nuclear reactor? Suggest other suitable materials for control rod and moderator.
2. a) Explain the working of ITER reactor with neat diagram.
b) What are the effects of nuclear radiations and what are its preventions?
3. a) Define the following terms related to solar radiation.
 - i) Diffuse radiation
 - ii) Beam radiation
 - iii) Insolation
 - iv) Solar Constant
b) Describe in brief, solar thermal energy storage and utilisation.

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4.
 - a) With solar energy utilization we can overcome to some extent power crisis due to fossil fuels please comment.
 - b) Write notes on various applications of wind energy.
5.
 - a) Write two methods of preparation of hydrogen on industrial scale. Discuss various methods of storage of hydrogen.
 - b) Explain the principle of electricity generation using hydrogen.
6.
 - a) List and explain the main constituents of biomass materials.
 - b) State various routes of biomass energy conversion to other form of energy.
7.
 - a) Explain direct and indirect methods of biomass resource utilization.
 - b) How is biomass produced? Explain biomass general chemical thermodynamics.
8. Write a short note on the following.
 - a) Passive heating system
 - b) Biomass classification
 - c) Fuel cell efficiencies
